

Optimization

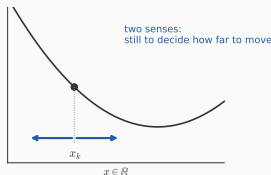
3.1 Multivariate optimization: definition and general remarks

José Firmino Aguilar Madeira

From Chapter 2 to 3: what has changed?

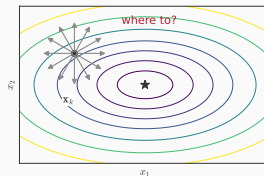
Chapter 3. Unconstrained multivariate optimization

Ch. 2: $x \in \mathbb{R}$



In 1D, the choice of direction reduces to **two options**, $\leftarrow$ and $\rightarrow$; once one is chosen (or the search interval), it remains to decide **how far to move**.

Ch. 3: $\mathbf{x} = (x_1, \dots, x_n)^T \in \mathbb{R}^n$

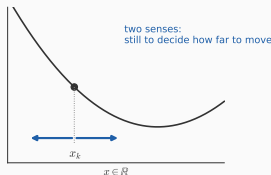


In n D, there are **infinitely many** possible directions from $\mathbf{x}_k$.

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Chapter 3. Unconstrained multivariate optimization

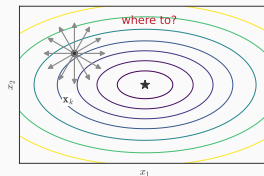
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1st question: where should we move?

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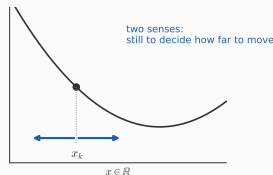


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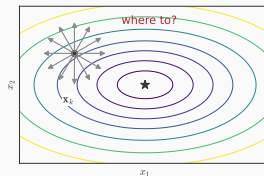


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1st question: where should we move?

2nd question: how far should we move in that direction?

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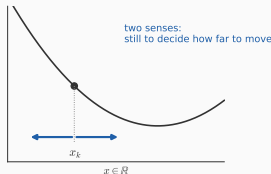


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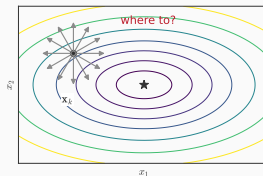
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In n D, there are **infinitely many** possible directions from $\mathbf{x}_k$.

1st question: where should we move?

2nd question: how far should we move in that direction?

$\mathbf{S}_k = \text{direction}$

$\lambda_k > 0 = \text{step}$

$\Rightarrow$

$\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{S}_k$

The multivariate problem: direction and step

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}), \quad \mathbf{x} = (x_1, \dots, x_n)^T,$$

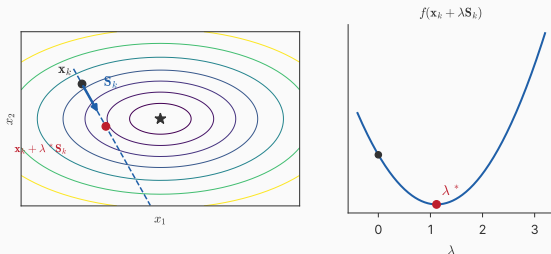
unconstrained (constraints are left for Ch. 4).

Maximizing f is minimizing $-f$.

The methods of Ch. 2 do not disappear

Once $\mathbf{S}_k$ is chosen, the step solves a **one-dimensional** problem:

$\min_{\lambda > 0} \varphi(\lambda) = f(\mathbf{x}_k + \lambda \mathbf{S}_k)$ — the **line search** (golden section, Powell, Newton...).



Ch. 3 chooses the **direction**

Ch. 2 can choose the **step**

This is the form of direction-based methods (3.3; in part, Hooke–Jeeves); random search, Nelder–Mead and Box progress in a different way (3.2).

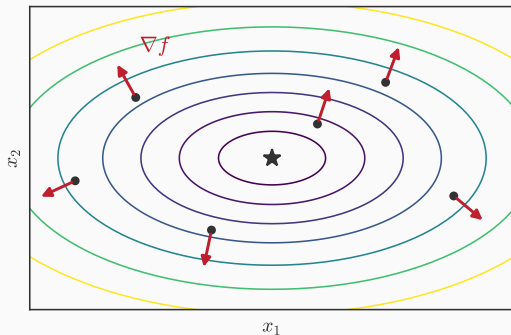
Gradient: where does f increase fastest?

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix} \quad (\text{R})$$

Along a unit direction $\mathbf{S}$, f changes at the rate $\nabla f(\mathbf{x})^T \mathbf{S} \leq \|\nabla f(\mathbf{x})\|$ (Cauchy–Schwarz), with equality for $\mathbf{S} = \nabla f / \|\nabla f\|$.

∇f is normal to the level curve

∇f points to the greatest local increase



$f(\mathbf{x}) = \frac{1}{2}x_1^2 + 2x_2^2$: the gradient at six points.

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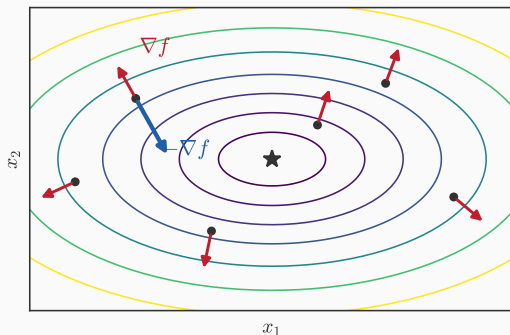
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To go down? The opposite direction: $-\nabla f(\mathbf{x})$

(This is the idea of the steepest descent method, 3.3.1.)



$f(\mathbf{x}) = \frac{1}{2}x_1^2 + 2x_2^2$: the gradient at six points.

What is a descent direction?

First-order Taylor (f differentiable), for $\lambda > 0$:

$$f(\mathbf{x} + \lambda \mathbf{S}) = f(\mathbf{x}) + \lambda \nabla f(\mathbf{x})^T \mathbf{S} + o(\lambda),$$

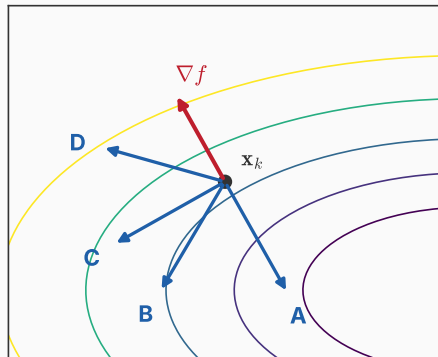
with $o(\lambda)/\lambda \rightarrow 0$ as $\lambda \rightarrow 0^+$.

If $\nabla f(\mathbf{x})^T \mathbf{S} < 0$, the linear term dominates for small λ : $f(\mathbf{x} + \lambda \mathbf{S}) < f(\mathbf{x})$.

Descent direction

$\mathbf{S}$ is a descent direction at $\mathbf{x}$ if $\nabla f(\mathbf{x})^T \mathbf{S} < 0$:
the angle between $\mathbf{S}$ and ∇f is **greater than 90°** .

Question: which of these arrows is a descent direction at $\mathbf{x}_k$?



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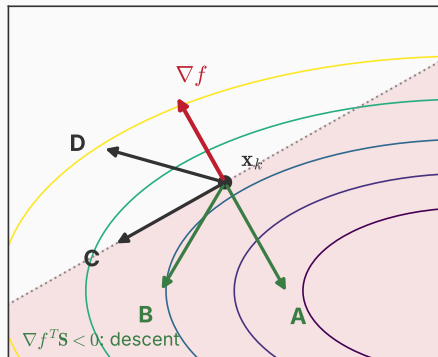
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A ($-\nabla f$) and **B**: yes. **C**: tangent to the level curve, $\nabla f^T \mathbf{S} = 0$ — *not* a descent direction (first order does not decide). **D**: $\nabla f^T \mathbf{S} > 0$, goes up.



And when we can no longer go down?

Let $\mathbf{x}^*$ be a local minimum (unconstrained). If $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$, the direction $\mathbf{S} = -\nabla f(\mathbf{x}^*)$ would give

$$\nabla f(\mathbf{x}^*)^T \mathbf{S} = -\|\nabla f(\mathbf{x}^*)\|^2 < 0,$$

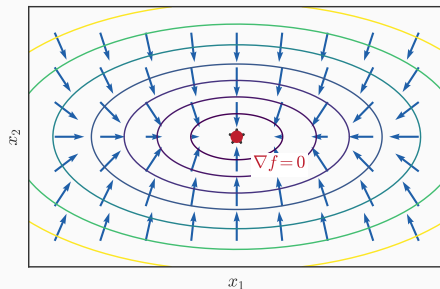
that is, a descent direction — and $\mathbf{x}^*$ would not be a minimum. Hence:

First-order necessary condition

f differentiable at $\mathbf{x}^*$ and $\mathbf{x}^*$ local minimum
 $\implies \nabla f(\mathbf{x}^*) = \mathbf{0}$.

Points with $\nabla f = \mathbf{0}$ are called **stationary points**.

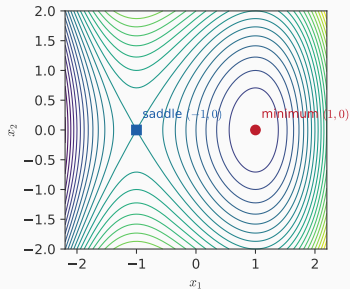
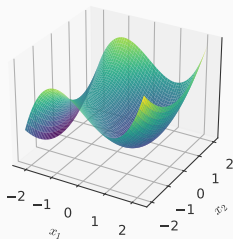
This is why $\|\nabla f(\mathbf{x}_k)\| \leq \varepsilon_g$ is a stopping criterion. With constraints (Ch. 4) this changes: at the minimum there may be descent directions that leave the feasible region.



Arrows: $-\nabla f$. Away from the minimum there is always a way down; at the minimum, there is not.

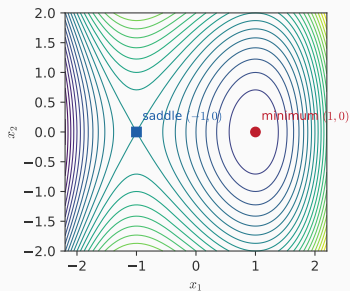
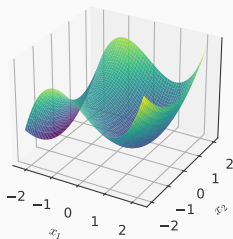
But stationary $\neq$ minimum

$$f(\mathbf{x}) = x_1^3 - 3x_1 + x_2^2: \quad \nabla f = \begin{pmatrix} 3x_1^2 - 3 \\ 2x_2 \end{pmatrix} = \mathbf{0} \iff \mathbf{x} \in \{(1, 0), (-1, 0)\}.$$



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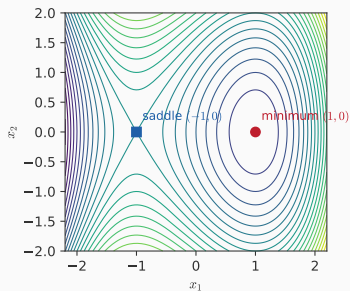
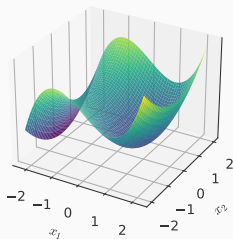
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Visually: $(1, 0)$ is a **minimum**; $(-1, 0)$ is a **saddle point** — it goes up along x_2 , down along x_1 .

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Visually: $(1, 0)$ is a **minimum**; $(-1, 0)$ is a **saddle point** — it goes up along x_2 , down along x_1 . **How do we tell them apart?** The gradient no longer helps (it is zero at both): we need the **curvature** $\Rightarrow$ the Hessian.

Hessian: the curvature

$$\nabla^2 f(\mathbf{x}) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{i,j=1}^n$$

symmetric if $f \in C^2(\mathbb{R})$.

Second-order Taylor:

$$\begin{aligned} f(\mathbf{x}^* + \mathbf{s}) &\approx f(\mathbf{x}^*) + \nabla f(\mathbf{x}^*)^T \mathbf{s} \\ &\quad + \frac{1}{2} \mathbf{s}^T \nabla^2 f(\mathbf{x}^*) \mathbf{s}. \end{aligned}$$

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At a stationary point the linear term vanishes:

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The **sign** of this quantity, for *all* directions $\mathbf{s}$, decides the local shape.

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$\nabla^2 f(\mathbf{x}^*)$	curvature	$\mathbf{x}^*$ is
pos. definite (eig. > 0)	upwards in all directions	strict local minimum
neg. definite (eig. < 0)	downwards in all directions	strict local maximum
indefinite (mixed signs)	up in one direction, down in another	saddle point
pos. semidefinite ($\lambda_{\min} = 0 < \lambda_{\max}$)	zero in some direction	not a maximum (inconclusive as to minimum)
all eigenvalues zero	zero in all directions	inconclusive

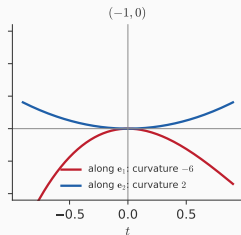
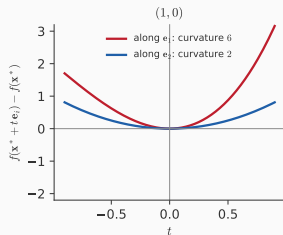
Second-order conditions ($f \in C^2$). Necessary: local minimum $\Rightarrow \nabla^2 f(\mathbf{x}^*)$ pos. semidefinite. Sufficient: $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\nabla^2 f(\mathbf{x}^*)$ pos. definite $\Rightarrow$ strict local minimum.
To check: eigenvalues or Sylvester (R).

Complete example: minimum or saddle?

$$f(\mathbf{x}) = x_1^3 - 3x_1 + x_2^2, \quad \nabla^2 f(\mathbf{x}) = \begin{pmatrix} 6x_1 & 0 \\ 0 & 2 \end{pmatrix}.$$

- $(1, 0)$: $\begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$, eigenvalues $6, 2 > 0$
 $\Rightarrow$ **strict local minimum**, $f = -2$.
- $(-1, 0)$: $\begin{pmatrix} -6 & 0 \\ 0 & 2 \end{pmatrix}$, eigenvalues $-6, 2$
 $\Rightarrow$ **saddle point**, $f = 2$.

In the slices on the right, the curvature of each slice is the diagonal entry $(\mathbf{e}_i^T \nabla^2 f \mathbf{e}_i)$.

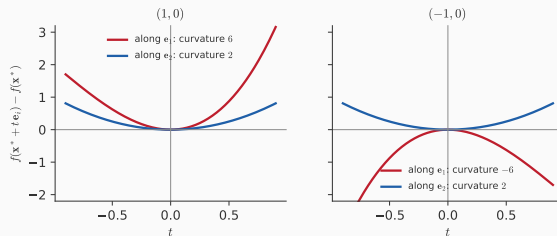


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There is no global minimum: $f(x_1, 0) \rightarrow -\infty$ as $x_1 \rightarrow -\infty$.

Optimality conditions are **local**: they say what happens in a neighbourhood, not on the whole domain. Seeking the *global* minimum is another problem (Ch. 5).

So how do we choose S_k ? The map of methods

Derivative-free (3.2) — only values $f(\mathbf{x})$

3.2.1 random search; 3.2.2 grid search;
3.2.3 Nelder–Mead; 3.2.4 Box; 3.2.5 Hooke–Jeeves.

They make progress by comparing values of f (which does not automatically make them robust to noise, 3.2).

Derivative-based (3.3) — ∇f , possibly $\nabla^2 f$

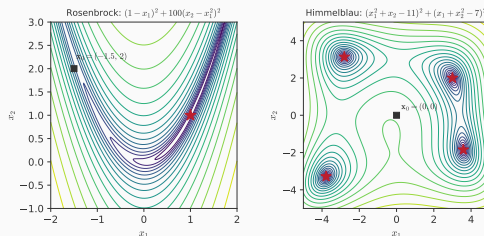
3.3.1 steepest descent: $\mathbf{S} = -\nabla f$
3.3.2 Newton: $\mathbf{S} = -(\nabla^2 f)^{-1} \nabla f$
3.3.3 conjugate gradients: $-\nabla f$ corrected by the previous direction

How much does the information cost? By central differences (2.6), a gradient costs $2n$ evaluations of f (forward: n , besides $f(\mathbf{x})$) and a Hessian $O(n^2)$.

What information do we have about f ? How much does it cost to obtain? What can we guarantee?

For each method: what it uses, how it chooses the direction, how much it costs (n_f , n_g , n_H , counted separately, 2.1) and how it behaves in the Rosenbrock valley. At the end (3.3.3), all seven on the same problem, with the same starting point and the same criterion.

Rosenbrock and Himmelblau: test tracks



Rosenbrock, $f(\mathbf{x}) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$: a single minimum, $(1, 1)$, at the bottom of a **narrow, curved** valley — it punishes methods that move along the axes or along the gradient.

Himmelblau, $f(\mathbf{x}) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$: four global minima, all with $f = 0$ — the result depends on the starting point.

They are not examples to “solve”: they are **test tracks** to see how the algorithms behave.

KEY TAKEAWAYS — 3.1

1. In direction-based methods, each iteration involves **two decisions**: the direction $\mathbf{S}_k$ and the step λ_k ; Ch. 2 can choose the step.
2. $\mathbf{S}$ is a **descent direction** if $\nabla f(\mathbf{x})^T \mathbf{S} < 0$; if $\nabla f(\mathbf{x}) \neq \mathbf{0}$, $-\nabla f(\mathbf{x})$ is always one.
3. At a local minimum (unconstrained, f differentiable), $\nabla f(\mathbf{x}^*) = \mathbf{0}$ — but this only identifies **stationary points**.
4. At a stationary point, the **Hessian** classifies locally: positive definite $\Rightarrow$ strict local minimum; indefinite $\Rightarrow$ saddle.
5. The choice of method depends on the **available information**: without derivatives $\rightarrow$ 3.2; with derivatives $\rightarrow$ 3.3.

Next (3.2): **if we do not know ∇f , how do we choose where to move?**

References Full list: *Course bibliography*, at the end of the course PDF.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 2.
- Dennis, J. E., Jr., Schnabel, R. B., *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, SIAM (Classics in Applied Mathematics 16), 1996.

Resources: —